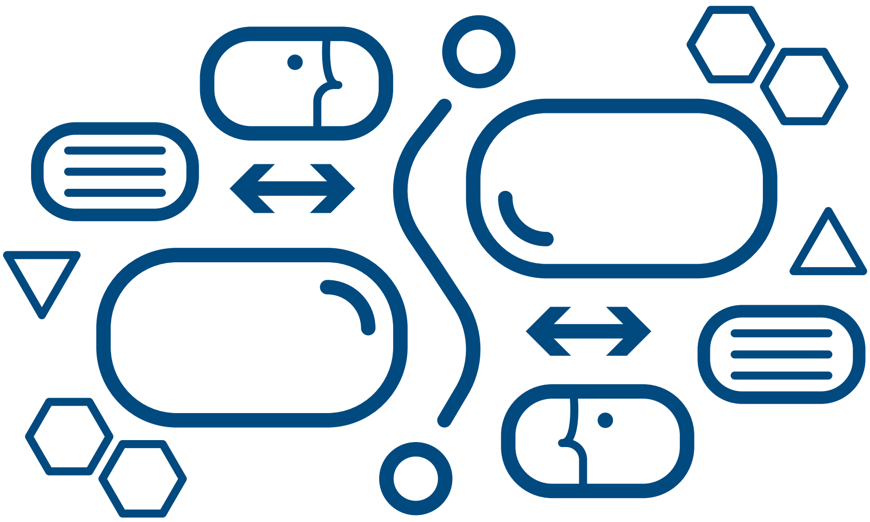
FROM OPEN DATA TO AI-READY DATA:

INTERNATIONAL PRACTICES FOR EMPOWERING DATA STEWARDS AND ADVANCING CITIZEN-GOVERNMENT DIALOGUE



OpenCulture Foundation

FRIEDRICH NAUMANN FOUNDATION For Freedom.

Taiwan / Global Innovation Hub

Analysis

Imprint

This research report was initiated by the Open Culture Foundation (OCF), Taiwan, and supported the Taiwan Office / Global Innovation Hub, Friedrich Naumann Foundation for Freedom (FNF).

We sincerely thank all interviewees, both domestic and international, for sharing the implementation details and practical experiences behind each case study. We also extend our gratitude to local contributors, whose hands-on experience across government agencies and civic communities helped clarify research limitations, refine analytical directions, and provide a concrete foundation for OCF's future advocacy and action.

Project Title From Open Data to AI-Ready Data: International Practices for Empowering Data Stewards and Advancing Citizen-Government Dialogue

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For citation, please use the following format:
Open Culture Foundation; Taiwan Office/Global Innovation Hub, Friedrich Naumann Foundation for Freedom (2026). From Open Data to AI-Ready Data: International Practices for Empowering Data Stewards and Advancing Citizen-Government Dialogue

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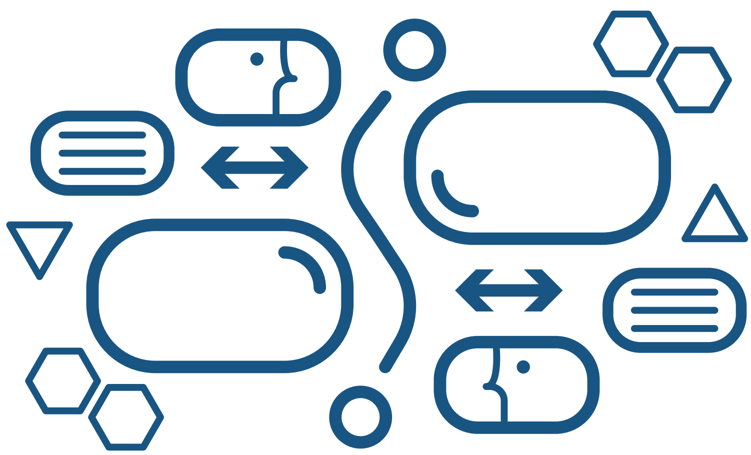


Executive Summary

The rapid evolution of artificial intelligence (AI) necessitates a fundamental paradigm shift from traditional open data to AI-ready open data. However, the structure and formats of existing open data from government do not necessarily fully align with the requirements of AI training and applications. To empower the government and citizens to transform traditional open data into AI-ready data, this report examines international case studies—spanning Spain, Germany, Cambodia, and France—to establish a practical mode of operation that redesigns the release formats of government data to encourage more application of such data in AI innovation. The following are the primary takeaways from the paper:

0 Government open data is essential to AI model developers because it is with lower risk of privacy and copyrights concerns and accessible. However, according to McKinsey & Company's report, many LLM researchers still spend around 80% of their time converting government open data into formats suitable for AI training, which shows the government needs to be empowered to transform open data to AI-ready one.

1 Transforming government's data release approach from legacy document formats to machine-readable standards and rigorous data architectures is the first and the most essential step for all nations to embrace AI-ready open data. This conversion significantly alleviates the technical burdens that currently slow national technological development and impose heavy processing costs on enterprises.

2 The establishment of a legal framework pertaining to open data is insufficient. In order to achieve AI-ready data, it is vital that governments improve or establish clear and comprehensive legal frameworks on data protection and intellectual property rights. Furthermore, it is essential for governments to orchestrate these frameworks, as their conflicts frequently impede the trajectory towards AI-ready data.

3 Furthermore, adopting international open licensing standards is critical for navigating complex regulatory landscapes, thereby reducing institutional risk and transaction costs for developers. To ensure data persistence and reliability, agencies must transition to proactive disclosure through centralized platforms and adopt resilient, multi-tiered storage strategies that integrate on-premise infrastructure with public repositories.

4 In order to persuade the government to transform data into AI-ready data or simply to open up more data, it is necessary to build trust in a number of ways. In the absence of an existing open data legal framework, advocacy groups initiate their engagement with the government by understanding its needs and selecting a less sensitive topic. This approach will facilitate the initiation of a dialogue with the government and provide a foundation for future collaboration. Furthermore, in instances where public servants exhibit reluctance, it is advisable to initiate action and demonstrate a willingness to assist them in resolving their issues. For instance, CityLAB's workshop for public servants invited them to voice their problems and thereby co-create AI tools to address them.

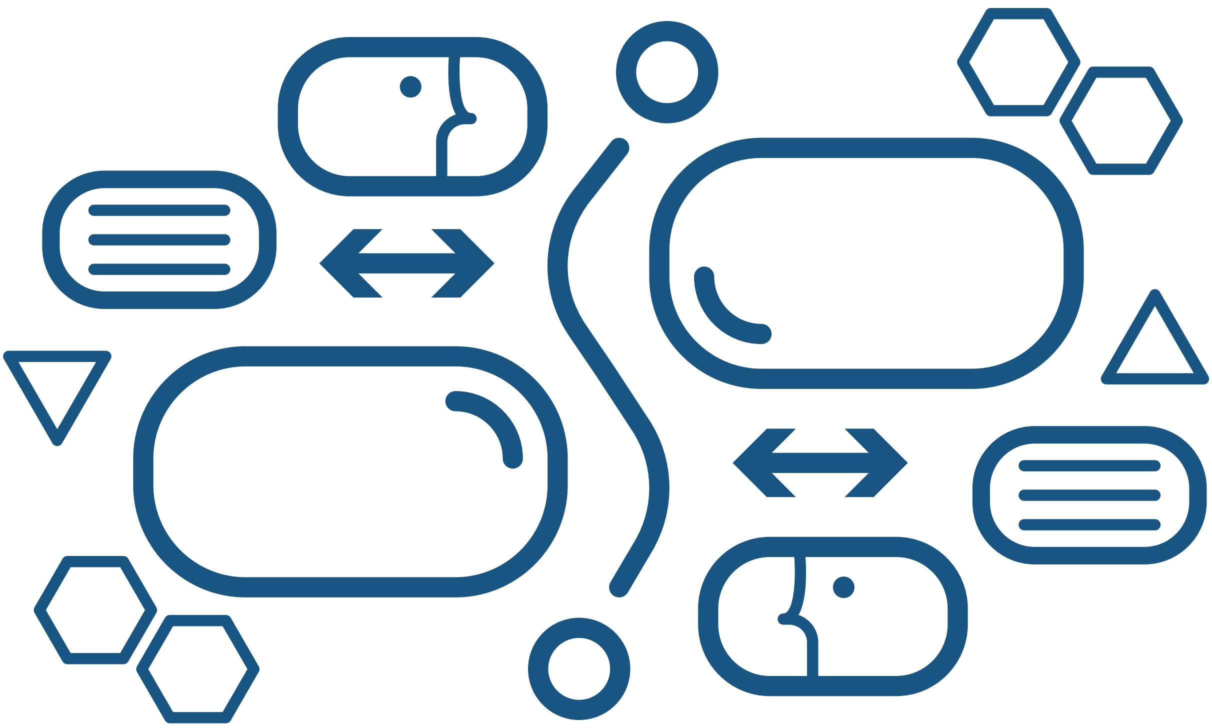
5 AI-readiness serves as the foundational infrastructure for broader societal and democratic empowerment. For civil servants, optimized governance improves internal administrative efficiency, transforming data release from a discretionary, reactive task into a core institutional function. For citizens, structured and interoperable data enhances civic agency by enabling the analysis of complex government records using AI tools, democratizing oversight regardless of individual technical resources.

6 Even though more technology tools can help us to transform government open data in pdf files into machine readable one, experience shows that it will still lose a significant amount of information, particularly the original structure or logic of a paper or articles. Therefore, it is still important that the government should release data with machine readable format. CVS is ideal for files such as numbers or statistics, and Markdown (MD) are ideal for text files.

7 In light of the inherent risk of model minorisation posed by AI, we must exercise greater caution to ensure that no personal or privacy data is included in the data corpus. This is because it will be difficult to remove such data from the AI model. Regardless of the government's decision regarding the release of the data, it is recommended that a test be conducted to ascertain the presence of any personal data. Should developers have concerns about the removal of personal data potentially disrupting the flow of data and, consequently, hampering the training of AI models, they may consider utilising a fixed pattern of words that can maintain the meaning of the sentences while replacing sensitive information.

8 Culturally, the development of sovereign AI corpora is essential to mitigate linguistic marginalization and temporal biases within global models. High-quality local data ensures that national contexts are accurately preserved, preventing the dominance of external cultural frameworks and ensuring diverse representation in the digital age.

In summary, this report positions AI-ready data as a collective public infrastructure project, dependent on sustained partnership between government, academia, and civil society. In order to achieve it, this paper also offers operable modules and roadmaps. The ultimate goal is to establish a resilient open data ecosystem in the age of AI.



Dispeling Common Myths about Making Open Data to AI Ready

Frequently Asked Questions

Reality / In Practice

Do agencies need to adopt new tools to release AI-Ready open data?

No. Agencies can follow their existing workflows to publish open data, directly releasing original format files such as documents and spreadsheets (e.g., .odt).

Will additional AI-Ready data requirements increase workload for agencies?

With proper process design, open data can already meet AI-Ready standards without requiring additional IT resources.

Will data need to be updated more frequently?

No. The "timeliness" requirement for AI-Ready open data focuses on clearly indicating when the data was produced, rather than updating all data in real time.

Chapter 1: Introduction

This report aims to introduce what AI-ready data is, assess its contributions to innovation and democratic governance, explore how it can be achieved, and how it is being practiced in countries around the world. By synthesizing these international experiences, the study provides a framework of regulatory and practical approaches to facilitate the creation of high-quality, AI-ready data. This report also demonstrates to the public sector officials that producing AI-ready data can often be integrated into existing open data workflows and, in many cases, may not require proprietary software; however, resourcing and staffing needs will vary by dataset and agency. In many cases, the main changes involve selecting appropriate formats and documenting update and temporal metadata (e.g., update frequency). This report also demonstrates to the public sector.

In order to encourage public agencies in different countries to proactively disseminate high-quality data and to increase the global volume of cross-cultural and cross-national AI-ready data— thereby fostering cultural diversity and the richness of human civilization—this study does not seek to provide narrow definitions of AI-ready data, open data, or AI models. Instead, it seeks to discussion which types of open data can be utilized within AI systems in a lawful and compliant manner, consistent with prevailing legal and regulatory frameworks.

At the same time, for anyone involved in civic technology or open-source communities, this study also welcomes you to, after reading the public–private collaboration processes around AI-ready data in different countries introduced in this research, draw on those experiences, tools used, or modes of cooperation, and use this research as a technical resource to support engagement and advocacy with government authorities.

1.1 Why This Case Study Is Needed

Since 2022, AI technologies have developed rapidly, and large language models (LLMs) have shown a continuously growing demand for training data. Due to their public nature and clear licensing, government open data has long been an important source for AI training. At the same time, countries around the world regard AI as a key driver of competitiveness, creating a synergistic ecosystem between open data initiatives and AI development.

However, the structure and formats of existing open data do not necessarily fully align with the requirements of AI training and applications. The optimization of data release standards to meet AI structural requirements—thereby streamlining training pipelines—has therefore become a priority for governments and public-sector decision-makers. This study examines international case studies to identify feasible approaches for upgrading existing open data into AI-ready data.

1.1.1 How Traditional Open Data Can Be Made AI-Usable

When comparing AI-ready data with traditional government open data, and taking the one-star level of Tim Berners-Lee’s (2006) 5-star Open Data rating system as a baseline, it becomes clear that both are fundamentally grounded in “open” licensing. However, because different AI systems have different training requirements, they require data in different formats—particularly formats and specifications that differ from those commonly found in basic open data [1]. Taking large language model (LLM) training as an example, most datasets released by government agencies on Taiwan’s open data platform have historically been provided in CSV format. Yet, because many field names and metadata are inconsistently defined or under-specified, this often leads to semantic ambiguity when models interpret the data during training. By contrast, the Markdown (MD) format, which is often referred to as the “native language” of LLMs, facilitates the structured processing of extensive textual corpora, such as reports and official documentation. Despite its efficiency, MD has not yet been widely adopted by government agencies in Taiwan. Similarly, although producing four- and five-star open data poses a higher threshold for public institutions, the defining features of these higher-rated data formats—such as persistent URIs and semantically linked data networks—provide critical technical foundations for the development of scalable AI-driven applications.

The same principle applies to multimodal AI. Images, audio, geospatial information, and other multimedia content are also essential for developing sovereign AI that can embody richer cultural contexts. In this regard, the construction of high-quality metadata is a core component of sovereign AI development. In the past, many valuable cultural, historical, or governmental report materials in Taiwan were preserved primarily in PDF format. To comply with open data regulations, public agencies have often sought to convert these files into other machine-readable formats. However, layout-driven design elements—such as embedded non-machine-readable images, mixed tables, and multi-column layouts—can make automated interpretation difficult. In recent years, advances in AI technologies, open-source tools for optical character recognition (OCR) and automated layout analysis have made it more feasible to integrate these digital assets into training pipelines. If these efforts are further complemented by more robust metadata—specifically through the precise annotation of document hierarchies and tables of contents—the accuracy of automated parsing can be significantly improved, thereby enhancing the availability of high-fidelity datasets for AI development.

1.1.2 Understanding “AI-Ready Data”

Against this background, the central concern of this study is how existing open data can be progressively improved into AI-ready open data. In Taiwan, one important point of reference in this process has been the FAIR data principles, which emphasize Findability, Accessibility, Interoperability, and Reusability [2]. These four dimensions, together with metadata governance frameworks as a practical entry point, provide a basis for strengthening data provenance transparency and improving overall data quality. The U.S. Department of Commerce likewise notes in *Generative AI and Open Data: Guidelines and Best Practices* that the needs of AI systems[2] for automated extraction and interpretation should be considered at the point of data release, so that published data already satisfies the baseline conditions needed to support AI training (U.S. Department of Commerce, 2025).

A related framework is the World Bank’s concept of AI-ready development data, which identifies three core conditions for advancing data toward AI readiness. The first is the development of AI-ready Data Systems, including data discovery platforms, application programming interfaces (APIs), and relevant technical standards that enable data to be effectively found, accessed, and used across systems. The second is the Provision of High-Quality Data and Metadata: data should be reliable and timely, and accompanied by structured, sufficiently complete metadata so that both machines and human users can accurately interpret its content and context. The third is Sound Governance and Strategic Collaboration, through which policy frameworks, standardized processes, and cross-sector coordination can help ensure data quality and consistency, enhance transparency, and promote the responsible use of data, thereby strengthening trust in the broader data ecosystem [].

Taken together, these principles suggest that AI-ready open data should be understood as data that is prepared, at the point of release, takes into account its potential future application scenarios; adopts appropriate data formats; and simultaneously establishes the necessary metadata. Where appropriate, releases can be accompanied by clearer documentation, quality controls, and reuse permissions, so that—when used for AI training—data is more likely to be accurate, complete, reliable, and extensible.

1. Berners-Lee, T. (2006). Linked data: Design issues, *http://www.w3.org/DesignIssues/LinkedData.html.* Please also read its further explanation *https://5stardata.info/zh-TW/*

2. Ministry of Digital Affairs. (2025). AI-Ready data metadata framework and indicator guidelines (1st ed.). Ministry of Digital Affairs. *https://www-api.moda.gov.tw/File/Get/moda/zh-tw/ae1g6Uvj4fUa4a3*

3. World Bank. (2025, July 21). From open data to AI-ready data: Building the foundations for responsible AI in development. World Bank Blogs. https://blogs.worldbank.org/en/opendata/from-open-data-to-ai-ready-data--building-the-foundations-for-re

1.1.3 Key Considerations for AI-Ready Open Data

When online information involves copyright infringement or privacy disputes, the spread of such information can be curtailed by removing the original webpage or requesting search engines to de-index it, reducing its visibility over time when information and data are used for. However, training AI models, they may be learned and reflected in model parameters during training, deeply internalizing the information within complex neural structures. In existing studies, this is described as high data persistence of AI systems [4]. Therefore, a pre-training audit should be conducted to assess licensing status of scalable AI-driven applications.
and applicable exceptions and limitations (e.g., fair use where relevant) and screen whether personal and privacy information still exist in the data.

1.3 Concrete Benefits of AI-Ready Open Data

In 2009, a research team led by Professor Fei-Fei Li at Stanford University created the open ImageNet dataset, which covers more than 20,000 categories and contains over 14 million labeled images[5]. This dataset fundamentally transformed the development of AI-based image recognition: image classification accuracy increased from 71.8% to over 95% within seven years, and ImageNet went on to become a foundational training resource for applications such as medical imaging, autonomous driving, and facial recognition.[6] This example demonstrates that high-quality, well-annotated public datasets can catalyze technological breakthroughs across entire industries. It also suggests that AI-ready data can be expected to bring tangible benefits to the development of human civilization.

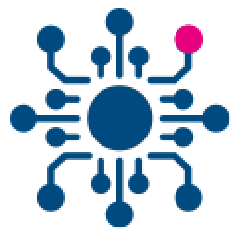
1.3.1 Industry Development and Value Creation

According to data from McKinsey & Company, open data can generate up to USD 5 trillion in economic value annually across seven major global industries. This enormous economic value is largely derived from the use of open data in AI training, highlighting the substantial industrial benefits of AI-ready data[7].

The report also notes that researchers working on LLM models spend as much as 80% of their time converting government open data into formats suitable for AI training. This underscores that if governments were able to produce AI-ready data directly, it would not only accelerate national technological development but also significantly reduce the high data processing costs borne by domestic enterprises.

1.3.2 Cultural Preservation and the Consolidation of Sovereignty

When a nation's linguistic data represents a marginal share of publicly available training datasets, its citizens are unlikely to receive information that accurately reflects their national context when using mainstream AI models. This can contribute to "marginalization" and "misinterpretation" over time, shaping persistent misunderstandings in public discourse and educational contexts. If unchecked, such conditions will undermine the preservation and accurate representation of cultural and historical knowledge within AI-mediated information environments, ultimately putting national identity and heritage at risk.



4. Schlarmann, Julian, and coauthors. "What Should LLMs Forget? Quantifying Personal Data in Large Language Models." arXiv, June 1, 2025, https://arxiv.org/abs/2507.11128.

5. Deng, Jia, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. "ImageNet: A Large-Scale Hierarchical Image Database." In 2009 IEEE Conference on Computer Vision and Pattern Recognition, 248–255. IEEE, 2009.

6. Blaivas, Michael, and Leslie Blaivas. "Are Convolutional Neural Networks Trained on ImageNet on Par with Humans for Point-of-Care Ultrasound Classification?" Journal of Ultrasound in Medicine 40, no. 6 (2021): 1099–1107.

7. Manyika, James, Michael Chui, Diana Farrell, Steve Van Kuiken, Peter Groves, and Elizabeth Almasi Doshi. "Open Data: Unlocking Innovation and Performance with Liquid Information." McKinsey & Company, October 2013, https://www.mckinsey.com/capabilities/tech-and-ai/our-insights/open-data-unlocking-innovation-and-performance-with-liquid-information.

Chapter 2: The Development of AI-Ready Open Data Around the World

This chapter first introduces how AI-ready open data is being implemented across diverse jurisdictions, along with case studies drawn from this research's interviews with several related projects and organizations. These examples aim to give readers a concrete understanding of the practical use of AI-ready data.

They also demonstrate how the pathways of achieving AI-ready data in different countires have been shaped by their different instituional contexts and policy objectives, and thereby making each county's implementation models various. This will help us to foresee different countries' policy trends and development in AI-ready data.

2.1 Global and Taiwan Development Trends

This section first compares four major trends in the development of AI-ready data worldwide and examines the specific trajectory adopted by Taiwan.

2.1.1 Development Paths of Countries Worldwide

The development of open data can be broadly categorized into three paths. The first one is scale- and market-oriented, exemplified by the United States, which leverages federal government data to drive AI applications and has issued related guidelines and standards.[8] The second path emphasizes regulation and trust; for example, the European Union ensures data release through legally binding directives and stresses the sustainable development of the AI economy under a robust regulatory framework. The third path focuses on digital sovereignty, as seen in Arabic-speaking countries, which carefully assess what data can be made public in order to protect local languages and cultures.

East Asian countries and Singapore largely adopt hybrid approaches. Japan, influenced by population aging and labor shortages, grants greater flexibility in data use to accelerate AI development. South Korea and Singapore are led by their governments in planning, treating data governance and AI as elements of national competitiveness, and promoting data circulation and AI applications through policies and institutional frameworks.

2.1.2 Recent Developments in Taiwan

Since 2012, Taiwan has promoted the opening of government data, and by the end of 2020, over 48,000 datasets had been released. In 2020, the "Digital Government Program 2.0" initiative was launched, emphasizing data quality and prioritizing high-value areas such as transportation, healthcare, and the environment. Following the establishment of the Ministry of Digital Affairs, Taiwan has launched the "Open Data 2.0" program, facilitating data structuring, API applications, and cross-domain collaboration to support AI training.

With the rise of generative AI, Taiwan has demonstrated a proactive commitment to constructing sovereign AI corpora. In 2023, the National Science and Technology Council launched TAIDE, integrating diverse local datasets to create large-scale traditional Chinese language models. The Ministry of Digital Affairs released a Beta version of a corpus in 2025, initially containing approximately 600 million tokens covering legal, cultural, educational, and other agency datasets, with a real-name licensing system to protect copyrights.
The government has also revised regulations to accelerate the release of non-sensitive data and is planning the "Regulation on Promoting the Development of Innovative Use of Data." In the private sector, organizations such as the Information Managers Association are promoting initiatives like "Taiwan Tongues," supplementing local corpora with literary assets and regional linguistic variations to enhance cultural diversity.

8. U.S. Department of Commerce, Generative Artificial Intelligence and Open Data: Guidelines and Best Practices, Version 1, January 16, 2025.

TOE Analysis Framework

	Key Focus Areas	Focus Areas and Themes in This Research
Technological Context	Focus on data-centered technological features and processing capabilities, as well as how these are currently implemented in practice in the case studies.	1. Data Processing Technical Workflow 2. Technical Details of Privacy Protection and De-identification Practices 3. Methods for Mitigating Bias
Organizational Context	Analyze the internal collaboration patterns of the implementing organizations and how collaboration is coordinated across different organizations.	1. Selecting Appropriate Open Data Owned by the Organization 2. Coordinating Data Acquisition Across Agencies 3. Stakeholder Enagement by the Data Management Unit 4. Cross-Agency Collaboration Models (Public-to-Public, Public-to-Private)
Environmental Context	Examine the external environment in which the organization operates, focusing mainly on the legal context and the forms of adaptation generated by institutional arrangements.	1. Applicable Legal and Regulatory Background 2. Compliance with Privacy Protection and De-identification Requirements 3. Compliance with Copyright and Licensing Terms

2.2 Case Studies from Countries Around the World

In addition to introducing global trends in AI-ready applications, this study also interviews practical agencies and project teams from different regions, helping readers gain a concrete understanding of public–private collaboration around AI-ready data, and analyzes these experiences in the following chapters.

2.2.1 Case Analysis Framework

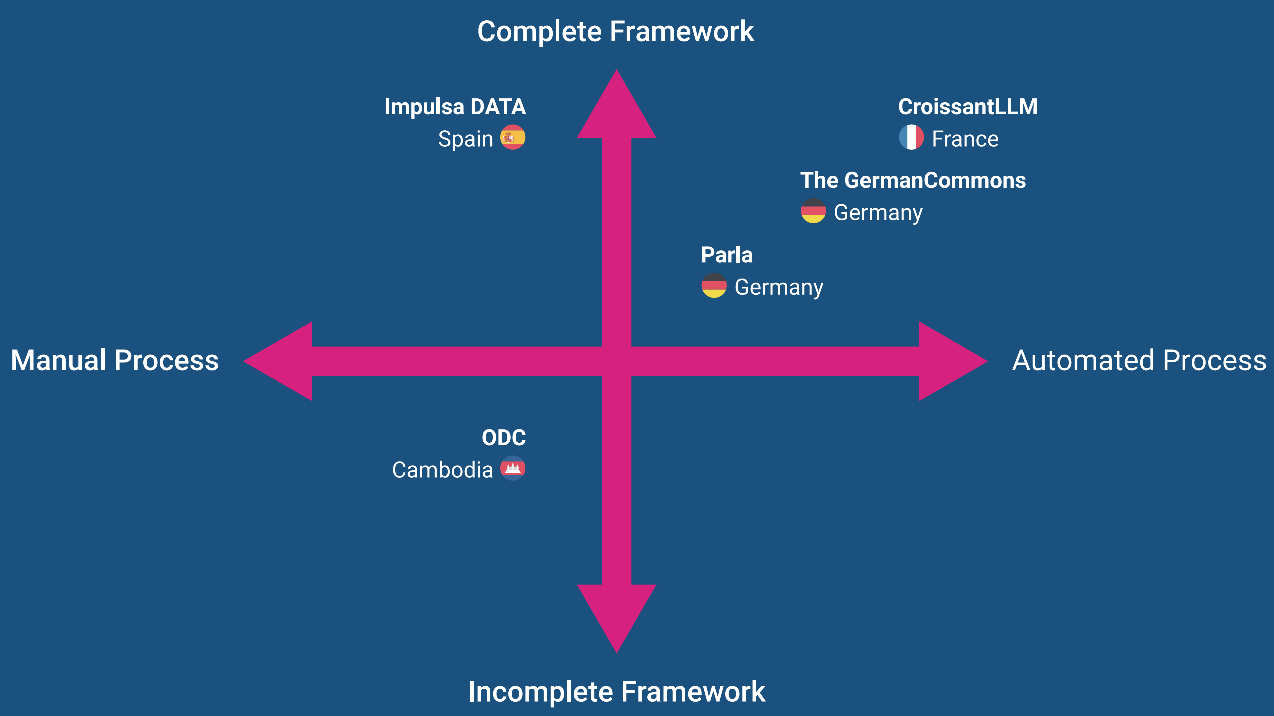
This study adopts the Technology-Organization-Environment (TOE) framework proposed by Tornatzky and Fleischer (1990) as an analytical tool[9] to evaluate cases and summarize interview insights. The TOE framework is a widely used theoretical model for studying the adoption and implementation of technological innovation. It examines the process through three dimensions: Technological Context, Organizational Context, and Environmental Context. It asks, "Under what kinds of social environments and organizational collaborations, and within what kinds of technological contexts, can more open data be released in order to develop a core focus on AI-ready open data?" and considers the strategies or approaches required across these different dimensions. The specific dimensions are shown in the table above.:

A more detailed analysis of the technological context will be presented in "Chapter Three: How Open Data Becomes AI-Ready," where each organization's technical capabilities and stages of technological development will be examined. The interactions between organizational and environmental contexts will be further explored in "Chapter Four: What Environments, Policies, and Adaptive Actions Are Needed to Advance AI-Ready Data."

2.2.2 Case Selection and Positioning

To select representative case studies from international examples, this study employs the TOE framework's three contextual dimensions to construct a dual-axis analytical matrix. During the literature review, identified cases were mapped onto this analytical matrix to ensure that cases across different levels of technological and institutional development, as well as varying organizational and legal contexts, are included for exploration.

9. Tornatzky, L. G., & Fleischer, M. (1990). The processes of technological innovation. Lexington,



Horizontal Axis (Technological Maturity) reflects differences in each case's data processing capabilities, ranging from heavily manual processes to large-scale automated processing;

Vertical Axis (Institutional and Organizational Completeness) combines the development level of legal frameworks and organizational collaboration, ranging from the absence of clear regulations and coordination structures to fully established and institutionalized governance frameworks (including laws, cross-departmental collaboration, and resource integration). The analysis is illustrated in the figure below:

2.3 Case Overview and Key Highlights

2.3.1 Spain's ImpulsaDATA Program

The ImpulsaDATA program is led by Carlos Alonso Peña,[10] Director of the Spanish Data Office Division, and aims to guide Spanish public institutions through the challenges of open data caused by the need to comply with multiple overlapping regulations.

ImpulsaDATA provides companion-style support, assisting ministries and agencies in identifying which data can be safely released when they are uncertain about how to reconcile overlapping regulatory mandates. The program addresses institutional concerns through a practical "diagnose–gap–act" methodology and applied operational collaboration frameworks.

Interviewees:
Directorate-General for Data at the Ministry for Digital Transformation and Public Administration
Carlos Alonso Peña

2.3.2 Germany's German Commons Project

The German Commons project is led by Lukas Gienapp and Martin Potthast,[11] and aims to provide more high-quality, openly licensed German-language training data. The project consolidates 41 data sources with clear licenses (such as CC-BY-SA 4.0) and also releases an automated code repository covering the full workflow, including data cleaning and removal of personally identifiable information. By adhering to current copyright regulations in its data processing workflow, the project seeks to support the development of a fully open German-language large language model.

Interviewees:
Co-Founder
Lukas Gienapp
Martin Potthast

10. datos.gob.es. "Impulsadata: Nuevo Servicio de Apoyo a la Apertura de Datos de Alto Valor." datos.gob.es, 2025, *https://datos.gob.es/es/noticias/impulsadata-nuevo-servicio-de-apoyo-la-apertura-de-datos-de-alto-valor.*

11. Gienapp, Lukas, et al. "The German Commons - 154 Billion Tokens of Openly Licensed Text for German Language Models." arXiv:2510.13996 [cs.CL], October 15, 2025, *https://arxiv.org/abs/2510.13996* .

Geographical Distribution of Case Studies

2.3.3 Cambodia's Open Development Cambodia

Open Development Cambodia (ODC) is a civil society organization with more than a decade[12] of operational experience. In a context where government transparency laws , access to information law, and personal data protection regulations are still underdeveloped, ODC leverages practical technical assistance to help the Cambodian government, as well as other stakeholders, demonstrating its advocating data sharing across relevant actors.

Through an approach centered on building trust and maintaining a neutral position, ODC successfully bridges the gaps in the legal framework for data openness. Over time, this trust has evolved into long-term collaboration, enhancing government agencies' data management and integration capabilities.

Interviewees:
Program and Strategy Associate **Wen-Ling (Amy) LAI**
Partnership and Program Manager **Vimoil OURN**
Executive Director/Editor-in-Chief **Try THY**

2.3.4 Germany's Parla Project

Parla was jointly developed by the Senate Chancellery of Berlin and CityLAB Berlin. The core goal of the project is to optimize the retrieval process for Berlin's public administration documents. Currently, the system can extract over 11,000 publicly available parliamentary documents from the website and automatically generate responses to citizens' inquiries in the form of a generative AI chatbot.

Interviewees:
Lead Prototyping at CityLAB Berlin **Ingo Hinterding**

2.3.5 France's CroissantLLM Project

The French CroissantLLM project leverages datasets released through France's government open data initiative platform, aggregating data from multiple ministries.[13] Using these openly licensed government datasets, the project trained a 1.3-billion-parameter, truly bilingual open language model. The initiative primarily utilizes cross-ministry datasets aggregated by the platform to develop a model with high transparency and performance.

Interviewees:
Co-Founder **Manuel Faysse**

12. Open Development Cambodia. "Home." Open Development Cambodia, accessed December 24, 2025, *https://opendevelopmentcambodia.net/*.

13. Faysse, Manuel, et al. "CroissantLLM: A Truly Bilingual French-English Language Model." arXiv:2402.00786 [cs.CL], February 1, 2024 (v1), last revised April 9, 2025 (v5), *https://arxiv.org/abs/2402.00786*.

This image was generated by using Google Gemini.

Chapter 3: How Open Data Becomes AI-Ready

This chapter summarizes the technical experiences of transforming open data into AI-ready data based on the case study interviews presented earlier. Many people assume that upgrading to AI-ready requires expensive software or labor-intensive manual intervention for data cleaning. However, the interviews indicate that in most cases, simply following existing open data workflows—even in a simplified form—can effectively produce AI-ready data.

A practical approach to AI-ready data prioritizes preserving information in its native, machine-readable form and avoiding conversions (e.g., scanned PDFs) that degrade structure; in many cases, this can be achieved with limited additional processing.

This chapter aims to help agencies optimize the production process for AI-ready open data.

Under the TOE analysis, this chapter focuses on the technological dimension, examining the key technologies required to advance open data toward AI-readiness. It uses case comparisons to illustrate three main topics: data format conversion and enhancement of machine readability, privacy protection and de-identification, and the identification of data bias.

3.1 Using Machine-Readable Formats

PDF document processing has become a common challenge for many teams due to modern reading habits. The Parla system developed by CityLAB Berlin aims to make 11,000 parliamentary documents searchable and summarizable by AI. However, because the files are in PDF format, they are not suitable for large language model processing and must first be converted to text using OCR, creating a significant technical debt challenge.

3.1.1 Systematic OCR Conversion and Document Preservation

The Parla team invested substantial resources in OCR to make the content machine-readable, but this caused the loss of structural information such as original formatting, paragraphs, and table headers, affecting semantic integrity and contextual coherence. They found that quality control required a hybrid of semi-automated processing and manual verification. At the start of the project, handling PDF files was critical, and the team had to undergo iterative cycles to ensure optimal outcomes. Eventually, they compiled their experience into technical documentation and shared it publicly to assist stakeholders in mitigating similar difficulties.

3.1.2 Addressing Readability Variations Across Historical Typefaces

The German Commons project faced similar challenges. The project consolidated German texts from 41 sources across seven domains, ultimately producing 154.5 billion tokens of training data. During the interview, project leader Lukas Gienapp explained that they lost nearly half of the original data during the data filtering stage. In addition to language-identification filtering, a major source of loss was errors introduced during OCR conversion.

The project included many older German Frakturschrift documents, which early OCR technology could not accurately recognize, resulting in numerous errors after digitization. To avoid undermining language model training, the team chose to strictly filter the data, accepting reduced dataset volume in order to maintain qualitative rigor. This illustrates that government records lacking machine-readable formats have limited utility for downstream AI training.

3.1.3 Selecting Appropriate OCR Strategies and Formats Based on Data Conditions

It is worth noting, however, that the German Commons team adopts two different technical approaches depending on the nature of the documents requiring OCR. For PDFs that already contain an embedded text layer, they use the tool Grobid[14] to extract text; for PDFs that are exclusively scanned images, they Allen Institute for AI's OlmOCR model[15] for recognition. This strategy of selecting tools based on document characteristics serves as a constructive precedent for other projects. Nevertheless, even with advanced tools like Grobid or Olmocr, current technological capabilities still have significant limitations in accurately recognizing older fonts or complex layouts.

3.1.4 Designing for Machine Readability and Open Licensing at the Point of Data Creation and Preservation

Based on the cases discussed above, several actionable insights can help stakeholders make open data more suitable for AI training. First, prioritize releasing data in machine-readable formats. Using conventional tabular data as a case in point: even if data is released in CSV format, merged cells or empty rows implemented for human legibility can still impede automated processing by AI systems. These "human-friendly" layout designs often become confusing noise during AI training. Therefore, it is recommended to also release raw files that are unformatted and maintain consistent header definitions.

If the nature of the data does not require spreadsheets, beyond traditional plain text files, Markdown format could also be considered. This format allows precise preservation of heading hierarchies, list structures, and references, all while maintaining a minimal storage footprint, enabling AI to understand the content while also capturing the logical structure of the document. Where cross-referential structural integrity is required, structured formats such as JSON or XML can provide a more rigorous data architecture, significantly reducing the cost of subsequent cleaning.

Furthermore, in the past, government agencies often saved files in PDF format—and sometimes even released them as PDFs—to prevent layout issues when opening documents on different operating systems or software. However, it is advisable to release the original data formats alongside PDFs whenever possible. For example, if the original data is a spreadsheet, release it directly as a CSV or Excel file; if it is a text document, release it as plain text or in a structured

14. GROBID. (n.d.). GROBID tools. Retrieved April 9, 2026, from https://grobid.org/
15. Allen Institute for AI. (n.d.). olmOCR. https://olmocr.allenai.org2

format such as JSON or XML. Furthermore, if PDFs must be provided for specific reasons, prefer PDFs with embedded text layers over purely scanned images. Finally, if institutional capacity allows, agencies could consider offering multiple formats simultaneously, enabling users with different needs to access the version most suitable for them.

France's CroissantLLM project demonstrates that releasing data in machine-readable formats from the outset can greatly enhance convenience and efficiency. The project trained a 1.3-billion-parameter bilingual model using open government data. All data were centralized on France's open data initiative platform (data.gouv.fr) and provided under permissive licenses such as the MIT License or Creative Commons, allowing users to download and use the data directly. The data were already standardized and passed basic validation checks, so the CroissantLLM team required only minimal cleaning before training. This case shows that a unified data platform, clear licensing, and pre-processing can effectively lower the data preparation costs for AI developers.

3.2 Personal Data De-Identification Process

Protecting personal data should be an integral part of the open data workflow and is a critical issue that all open data practitioners must address. When releasing raw data, the primary perceived risk for government agencies is unauthorized data disclosure.

3.2.1 Iterative Privacy Risk Mitigation within Controlled Processing Cycles

The CroissantLLM team, when training their French bilingual model, used government open data with permissive licenses and books already in the public domain, so de-identification was not necessary. These sources did not contain personal data, and preliminary administrative review had already removed most sensitive information. The team also stated in the interview that, because they used public government data without personally identifiable information (PII), no traditional de-identification procedures were applied.

Nevertheless, the team exercised caution. Before releasing the model, they conducted privacy verification by embedding specific strings in the training data to test whether the model would memorize and output sensitive content. The results showed that as long as a piece of data appeared fewer than four times in the corpus, the risk of disclosure was low. This indicates that memorization within language models is positively correlated with token frequency; low-frequency sensitive data exhibits a reduced probability of reconstruction. This research has been published at ICML and contributes to discussions on copyright issues for large language models.[16]

3.2.2 Context-Aware Design of De-identification and Anonymization Codes

For handling raw text, the German Commons approach provides a useful technical reference. Most of the data in this project came from historical documents, and the personal information they contained was already outdated. Project lead Lukas Gienapp explained that for historical data, most PII is over eighty years old and does not require urgent protection.

Nevertheless, German Commons still carried out PII removal because the dataset included a small amount of online-sourced text. They used Microsoft's Presidio framework along with German-optimized regular expressions (Regex) to detect and handle phone numbers, credit card numbers, addresses, bank account numbers, and other information that could be maliciously exploited. Technically, they added regex detection for Germany-specific formats, such as regional phone codes and specific bank account prefixes.

In the interview, Lukas Gienapp shared an important technical insight: when removing PII, sensitive information that has been detected should not be simply deleted but replaced with generic content. Traditional de-identification often removes sensitive strings entirely, but German Commons chooses to replace them with generic labels rather than deleting them. He illustrated this with an example: if all credit card numbers are simply removed, the trained model will never learn what a credit card number looks like; however, if a set of known safe, generic credit card numbers is used as replacements and randomly inserted in place of the original sensitive content, the model can still learn the relevant contextual patterns. Similarly, a sentence like "Zhang Junya lives in Taipei" would be replaced with "[PERSON] lives in [LOCATION]" rather than deleting names or locations. This preserves sentence structure, enabling the language model to learn grammar and context without producing broken sentences due to data removal.

In addition, Lukas Gienapp emphasized that this substitution process must be traceable. In the appendix of their paper, German Commons provides a detailed list of all the synthetic placeholders they used, enabling subsequent users to understand what modifications were made to the data and, if necessary, apply alternative replacement strategies.

3.2.3 Privacy Considerations for Marginalized and Vulnerable Communities

Beyond the technical dimension, ODC's experience illustrates how practitioners manage data dissemination within nascent regulatory landscapes, while also reminding readers to consider communal and cultural data sensitivities beyond the

16. Meeus, Matthieu, Igor Shilov, Manuel Faysse, and Yves-Alexandre de Montjoye. "Copyright Traps for Large Language Models." In Proceedings of the 41st International Conference on Machine Learning (ICML 2024), 2024. https://dl.acm.org/doi/10.5555/3692070.3693506.

law. In many developing countries or regions with incomplete legal systems, even data that is legally open may cause real harm if it involves the residences or identities of indigenous peoples or other vulnerable groups. ODC staff explained in the interview that, in building trust with government agencies, they learned to respect and communicate with their counterparts. For example, when collaborating with the Ministry of Environment, if the agency indicated that not all reports could be made public, ODC would ask regarding the rationale for non-disclosure, seek to understand their concerns, and learn to judge which reports could be released publicly and which should remain for research or educational purposes. At the same time, ODC would propose potential solutions whenever possible, facilitating dialogue with the government and other stakeholders.

Notably, ODC's approach highlights how culturally sensitive data should be handled in practice. In the interview, staff shared a critical analytical insight: in Cambodia, many indigenous people are reluctant to acknowledge their identities because they wish to avoid legal discrimination. Even merely mentioning a name can potentially harm local communities. The lesson for this study is that protecting personal data goes beyond technical de-identification—it requires a deep understanding of local cultural contexts. When handling such data, ODC implements community guidelines that are stricter than legal requirements, ensuring that the release of data does not cause adverse downstream impacts to vulnerable groups. This human-centered approach cannot rely solely on automated tools or statutory minimum requirements; it demands cultural sensitivity to the local context and serves as an important reference for all open data practitioners.

3.2.4 Summary

From these cases, several practical principles can be drawn. First, managing data at the point of origin is preferable to addressing issues afterward; by prioritizing the release of datasets inherently devoid of sensitive personal information, the workload for subsequent de-identification can be greatly reduced. Second, when de-identification is necessary, replacement rather than deletion should be used to preserve the integrity and usability of the text. Finally, the de-identification process should be transparent and traceable, allowing users to understand what modifications the data have undergone and recalibrate parameters if necessary.

3.3 Eliminating Bias in Training Data

Language model developers must address bias in training data. The quality of the data determines the perspective of the AI; if the data is biased, the model's output will also be affected. Interviews with various teams revealed diverse forms of bias and different strategies to address them.

3.3.1 Intentional Dataset Composition and the Release of Open Tools

The CroissantLLM project aimed to build a truly bilingual language model. At the time, most models claiming to support French still relied heavily on English data, resulting in poor performance in French. To address this issue, the project prioritized strategic linguistic allocation from the outset, allocating 40% English, 40% French, and about 20% code, ensuring a balanced corpus and improving the model's capability and stability in both bilingual and structured data tasks.

Beyond the model itself, the CroissantLLM team also released high-quality French-language datasets and established a dedicated French evaluation benchmark, FrenchBench. This allows subsequent researchers and model trainers to more systematically examine whether models exhibit disproportionate skewness toward English, and whether French-language performance is being sacrificed during training. It also helps safeguard that highly localized knowledge contexts—such as those in administration and law—can be correctly understood and generated by models in future development, rather than being forced through an English-centric mode of thinking before responding to France's institutional and policy questions.

This experience offers important insights for open data practitioners working with minority languages. Many minority languages have limited online resources, and passively ingesting existing materials risks entrenching or exacerbating linguistic disparities. The experience of CroissantLLM shows that language balance requires deliberate design of data composition, along with the creation of datasets and tools for subsequent research and evaluation. In this way, non-dominant languages can gain opportunities for development and validation in the era of AI and large language models.

3.3.2 Recognizing Temporal Bias in Data

The German Commons project faced another form of bias—an imbalance along the temporal dimension. The team observed an interesting phenomenon they called "nostalgia bias." Because German copyright law generally keeps works under copyright until a statutory term after the author's death, a large proportion of freely usable cultural works and newspaper archives are historical in nature. This creates what project leader Lukas Gienapp refers to as nostalgia bias:

the training data are filled with texts from decades or even over a hundred years ago, while content from the middle period—between the very recent and the very distant past—is largely missing. As a result, the trained model may sound outdated in tone and vocabulary, as if it were living in the early twentieth century.

Lukas Gienapp acknowledged that this problem is difficult to fully resolve under the current legal framework, since more recent cultural works remain under copyright protection. To mitigate the deficit in modern data, their strategy has been to integrate sources from different domains to attain a degree of structural parity. They actively seek texts such as court judgments and government gazettes—contemporary materials that are legally open—to counter temporal bias. Web-based texts tend to skew toward more recent content. Scientific literature also often gravitates toward more recent periods due to the open-access movement, while historical documents provide necessary temporal depth. Although a perfectly even temporal distribution is difficult, combining sources can reduce extreme temporal bias and ensure the dataset is not confined to a single period.

3.3.3 Supporting Multilingual Diversity

Cambodia’s ODC experience underscores the practical and governance complexities of working with linguistic diversity in AI-ready data. Interviewees noted that Cambodia is not a Khmer-only context: alongside Khmer, multiple minority communities—such as the Bunong (M’nong), Brao, Kuoy, Lao, Jarai, Kreung, Kavet, Tampuan, and Kachok—live across different provinces and use distinct Indigenous languages.

However, most available digital corpora remain concentrated in Khmer and tend to reflect urban perspectives; as a result, rural and Indigenous voices are systematically under-represented in the data pipeline and risk continued marginalization within the digital sphere. Because it is not possible to expeditiously produce large amounts of indigenous-language data, ODC has used English as a third language to preserve these perspectives. Although this approach is controversial, English datasets enable researchers who do not understand Khmer or indigenous languages to train AI models that represent more diverse Cambodian viewpoints. This also highlights the role that governments should play: releasing data of high public value but low commercial appeal—such as local gazetteers, dialect records, and information on rural infrastructure—thereby helping mitigate cultural bias in AI systems and supporting locally grounded AI capabilities.

3.3.4 Reducing Bias through Workflow and Process Design

In the context of reducing bias, these cases all converge on a unified strategic response: systemic biases require structural countermeasures. Lukas Gienapp of German Commons suggests a two-stage workflow. In the first stage, domain experts take the lead; in the second stage, data aggregators step in to adjust and harmonize the results. In operational terms, domain experts first curate specialized datasets within their own fields. Then, aggregators—who have expertise in corpus development, system architecture, and data processing—use technical methods to integrate these datasets. The advantage of this division of labor is that domain experts possess a more nuanced comprehension of the characteristics and potential biases of data in their fields, while aggregators can balance different sources from a cross-domain perspective.

3.3.5 Summary

These experiences lead to several immediately actionable recommendations. First, when formulating strategic data dissemination plans, agencies should review representativeness: consider whether certain groups or time periods are missing, and supplement them where possible. Second, if agencies have multilingual capacity, they should provide multiple language versions—the provision of translated executive summaries can improve accessibility. Finally, for fields with high professional barriers, explanatory documentation should be included to help users understand the data’ s context and limitations.

Eliminating bias does not mean achieving absolute algorithmic neutrality. Every dataset reflects the choices and constraints of its production process. The primary objective is to remain vigilant, mitigate bias where feasible, and disclose limitations transparently. German Commons offers a useful model by analyzing the temporal distribution and disciplinary composition of its data, allowing users to judge applicability against their specific use cases and requirements.

This image was generated using by Google Gemini.

Chapter 4: What Environments, Policies, and Adaptive Actions Are Essential to Advance AI-Ready Data

This chapter uses the TOE analysis framework to examine the cases collected in this study, highlighting the institutional, organizational, and environmental contexts that enable open data to gradually transition towards AI-interoperable frameworks. By comparing different countries and cases, the research team aims to show that there is no perfect institutional environment to acheive the goal of AI-ready data. Rather, different governance models can develop implementation approaches suited to their contexts. The chapter further examines how each case advances AI-ready data initiatives through institutional design and organizational collaboration within existing governance structures, regulatory regimes, and available policy support.

4.1 Legal Frameworks and Social Institutions for AI-Ready Data

This section focuses on the environmental context in the TOE analysis framework, examining how legal systems and social governance structures define the parameters for the evolution of open data into AI-ready assets. The environmental context considers not only the mere presence of regulations or institutions but also the broader social landscape, including the comprehensiveness of data-related legal concepts, public awareness and expectations regarding privacy and personal data protection, the implementation of copyright and licensing systems, and the social trust foundation for data use and AI development. Through case analyses, this section illustrates how different legal frameworks and social institutions influence the feasibility of data release, the operational space available to organizations, and the practical approaches used to promote AI-ready open data.

4.1.1 Clear Legal Frameworks for Open Data

4.1.1.1 Policies and Legislation Supporting Public Information Disclosure

The data acquisition processes of CroissantLLM and Parla clearly demonstrate that a well-established open data legal framework can reduce communication costs for developers and improve operational efficiency, paving the way for AI-driven research. CroissantLLM researcher Manuel Faysse noted in the interview that France's open data initiative significantly enhances the efficiency of data accessibility. Because the government has released a large volume of data under permissive licenses, team members openly stated that data licensing "was not a challenge for us; the model development process only required minimal cleaning of these open datasets."

Ingo Hinterding, the product lead for Parla, emphasized that "as long as the research team is handling publicly available data that does not require special protection, the development process is significantly accelerated." In some countries, citizens who wish to access government data must obtain formal written clearance from the relevant agency. In practice, however, citizens may not know which agency to contact and therefore cannot obtain the data. In rare cases, even if developers receive verbal permission from an agency, the uncertainty of such informal agreements can generate unforeseen institutional liabilities and operational risks.

4.1.1.2 Correctly Choosing Appropriate Licensing Terms

While a country's policies or legal framework are certainly important, in practice, the primary operational prerequisite lies in correctly labeling datasets with the "right" licensing terms. Each license type imposes different rules and restrictions on use, and these differences affect whether a dataset can be lawfully and practically included in AI training corpora. Common licensing terms specify whether the data can be used for commercial purposes, whether modifications or derivative works are allowed, or whether attribution is required. Over time, standardized licenses have emerged internationally, such as Creative Commons (CC) licenses and open government data licenses. Governments need to evaluate stakeholder requirements to designate licensing frameworks that provide the greatest benefit to the public.

In the case of France's CroissantLLM, it can be observed that the French government had already pre-labeled the licensing terms for datasets on the platform when releasing open data. This allowed the team to identify training-eligible datasets from a large portal based on the stated license conditions, thereby reducing avoidable legal uncertainty and rework.

4.1.1.3 Clear and Coherent Regulatory Framework

In the previous two sections, this study discussed the importance of licensing terms and supportive policies or legislation. Now we turn to the broader regulatory framework. When public agencies implement policies or adopt new legislation, failing to clarify relationships and potential conflicts among statutes can leave civil servants uncertain about how to proceed.

For example, public institutions in Spain must simultaneously comply with the Directive on the Re-Use of Public Sector Information (PSI Directive), the General Data Protection Regulation (GDPR), the Data Governance Act (DGA), the European Data Act, and the emerging AI Act. Without a clear hierarchy or prioritization among these laws, a "spaghetti bowl of compliance" situation arises. Regulatory ambiguity and perceived compliance risks can lead agencies to hesitate or even refuse to release data.

To address this issue, the European Commission has launched a Legislative Consolidation initiative. The EU aims to unify definitions across various regulations—for instance, clarifying how the GDPR's "legitimate interest"[17] lawful basis may be interpreted in AI training contexts—and to simplify reporting obligations by implementing a "report once" principle. These measures are intended to reduce barriers to promoting open data.

4.1.1.4 Summary

The cases demonstrate that combining an open data system with clearly defined licensing terms and supportive legislation helps research teams focus on AI development without being burdened by complex application procedures. A dataset that is clear and consistent is of higher utility than a large dataset with ambiguous terms. The overlapping regulatory issues in Spain also highlight that a lack of legal integration can cause public agencies to hesitate due to risk concerns.

17. Under the GDPR, "legitimate interest" permits personal-data processing when it is necessary for the legitimate interests pursued by the controller or a third party, provided those interests are not overridden by the data subject's interests or fundamental rights and freedoms—particularly where the data subject is a child.

4.1.2 Demonstrating the Value of Open Data to Promote Substantive Collaboration

Cambodia has not yet passed a law on government information disclosure or similar legislation granting citizens the legal right to request government data. Regarding personal data protection, although a bill has been discussed for nearly a decade, the legislative process has been slow due to negotiations and compromises among multiple stakeholders. In contrast, the country has already released a draft national AI strategy, identifying priority sectors such as education, agriculture, and manufacturing. However, documents about this strategy drafts remain at an early stage and have not yet been translated into concrete regulations or implementation guidelines. Despite this context, the civil society organization ODC has attempted to make breakthroughs, and its experiences offer valuable lessons.

4.1.2.1 Understanding the Need of Government

ODC employs a non-adversarial advocacy strategy, offering bespoke technical facilitation to each government agency to build trust and encourage voluntary data sharing. This voluntary, rather than mandatory, approach makes Cambodian government departments more willing to open data, as they do not perceive data requests as a threat to their authority.

For example, in the case of publicizing Economic Land Concessions data, ODC integrated disaggregated information from different departments to create comprehensive maps, helping define land boundaries and prevent conflicts. This work enabled relevant Cambodian agencies to clearly delineate each concession area, confirm boundaries with environmental protection zones, and avoid overlaps or disputes between different land uses. ODC' s intention was not primarily to retrieve restricted data, but to provide technical facilitation by consolidating and visualizing dispersed departmental information. During this coordination process, as the government recognized that cross-departmental integration could reduce decision-making errors and improve efficiency, previously restricted land information gradually became accessible. This model—starting from governmental needs to facilitate cross-agency information flow—not only addresses immediate issues but also transforms cleaned and structured spatial data into valuable local datasets for future AI models in land forecasting or resource planning.

4.1.2.2 Positive Public-Private Collaboration Cases Promote Legislation

Even more encouraging is that successful public-private collaboration can help drive legislation, filling gaps in national legal frameworks.

After the Economic Land Concessions data were made public in Cambodia, its impact extended further. Land designated as concession areas often encroached on indigenous communities, causing disputes. With map visualizations, indigenous groups could clearly see which lands were under threat and advocate for their protection with the government. For government agencies, these tools revealed policy blind spots, prompting departments to enhance land administration and optimize resource allocation. International investors and civil society organizations could also use the open data to conduct rigorous impact assessments regarding investment impacts. These experiences provide concrete evidence and justification for future legislation.

4.1.2.3 Summary

After understanding the benefits and operational practices of open data, policymakers became more confident in promoting legal reforms. According to the interview with ODC, the Cambodian government has recognized the problem of inconsistent data structures across departments and has begun discussions on establishing a unified governance framework. ODC's experience, including data format design, quality management, and balancing openness with protection, has provided valuable reference points for policymakers.

4.2 Comparative Analysis of Collaboration Models

This section focuses on the organizational context in the TOE analysis framework, examining the practical challenges faced in cross-organization and inter-agency collaboration during the promotion of open data and AI-ready initiatives. The organizational context emphasizes differences among participating units in organizational culture, roles and responsibilities, resource allocation, and decision-making processes, all of which directly affect whether data can be seamlessly released and reused.

In practice, different government agencies may adopt disparate institutional dispositions and approaches to open data due to differences in their operational goals, risk tolerance, and understanding of data openness. At the same time, when civil society or academic institutions participate in data initiatives, differences in organizational culture and work pace can also influence communication efficiency and collaborative outcomes. Therefore, establishing clear collaboration mechanisms and role definitions becomes a critical task in promoting open data and AI-ready data initiatives.

4.2.1 Establishing Institutionalized Inter-Agency Collaboration within Government

Berlin's Parla project established a long-term collaboration framework spanning both administrative and legislative branches, rather than merely a simple procurement relationship. Through regular innovation grants from the Berlin Senate, administrative agencies empowered CityLAB to act as a "bridge between information technology and government departments," enabling the innovation lab to operate cross-functionally and integrate parliamentary records with the operational data of various Senate offices. Essentially, this grant mechanism represents an investment in "collaborative infrastructure."

Spain's ImpulsaDATA initiative demonstrates another model of achieving inter-agency collaboration through institutionalization. The Spanish government established the Data Office as a centralized administrative node. Leveraging a statutory national data framework and annual reporting mechanisms, the office elevated simple inter-agency consultation into a legal obligation. Within the ImpulsaDATA framework, the Data Office does not merely provide advice; it standardizes a process—from diagnosis and gap analysis to concrete action—translating complex EU regulations into actionable plans for government agencies.

4.2.2 Integrating Needs through a Bottom-Up Approach

As previously noted in the case studies, CityLAB conducted "AI Ideas Workshop" sessions, inviting Berlin administrative staff to collaboratively discuss "which routine tasks could be supported by AI?" Participants highlighted that receiving and responding to parliamentary written inquiries was a time-consuming task that consumed significant resources. CityLAB therefore addressed this specific operational bottleneck directly—without needing additional persuasion for administrative adoption. This development model, which integrates user needs from the conceptual stage, ensured the project's practical relevance. A similar approach can be seen in Cambodia with ODC.

ODC provides technical support and training to government agencies willing to collaborate, helping them improve data management capabilities in a mutually beneficial way. This direct technical facilitation allows agencies to experience the benefits of cooperation, making them more willing to adopt a proactive approach to data release. For example, the Ministry of Post and Telecommunications and the Ministry of Environment were relatively open, inviting ODC to participate in policy draft discussions and, under certain conditions, share data; the Ministry of Planning also accepted technical training and support offered by the organization. Both cases demonstrate that, beyond technical development skills, the ability to understand practical needs and engage in cross-domain dialogue is crucial.

4.2.3 Selection of Data Topics

4.2.3.1 Consider the Complexity of the Data Topic

In general, different data topics imply that organizations must handle varying levels of complexity inherent in the data itself. Before discussing data selection strategies, the experience of the German Commons provides an important point of reference for this study: when facing vast potential data sources, systematic topic selection and strategic prioritization remain essential. Prior to deciding which sources to select, it is also necessary to gain a broad understanding of the existing data sources worldwide in order to choose the most representative datasets.

From the outset, the German Commons team established clear domain boundaries: law, science, culture, politics, news, economics, and web texts, ultimately collecting 154.56 billion tokens from 41 sources. While this seems like a massive amount, it is attributed to careful selection. The team explicitly stated that their strategy was to choose the "largest and most representative single source" in each domain, rather than attempting to include all available datasets.

However, it is equally important to understand what data sources exist globally at the start. For example, in the cultural domain, although German is Germany's primary language, high-quality German content does not exist solely within Germany. National libraries in Switzerland and Austria also hold rich German-language materials with open licenses. However, because the team only expanded their perspective to these cross-border sources later in the project, they had to undergo iterative refinements of their data processing workflows. This experience illustrates that at the initial stage, implementers should adopt a transnational perspective, conduct cross-jurisdictional assessments, and map all institutional data holders globally. This "comprehensive first, filter later" approach not only ensures the completeness of data coverage but also effectively avoids the sunk costs associated with retrospective data integration.

4.2.3.2 Emphasizing Data Applications to Lower Government Concerns about Releasing such Data

Although both the German Commons and ODC use government open data to produce AI-training datasets and pay close attention to topic selection, their purposes are entirely different. The German Commons' topic selection strategy is primarily driven by ensuring data quality, whereas ODC approaches topic selection from the perspective of facilitating access to government data, especially when open data involves areas such as fiscal revenue, mining rights, or land interests.

In the interview, ODC noted that datasets related to taxation or mineral extraction involve conflicting institutional and commercial interests, making government agencies more cautious or even conservative, often deflecting responsibility when faced with data requests. For example, when requesting mining revenue data, the Ministry of Mines and Energy would state that they do not hold the data and refer the request to the Ministry of Economy and Finance, while the Ministry of Economy and Finance would respond that they only have aggregate figures, not detailed company-level data. This institutional responsibility deflection reflects that when data involve sensitive commercial interests or government revenue, agencies tend to adopt the most conservative stance.

In response to this situation, ODC developed a strategic, gradual approach. For more conservative agencies, the organization does not apply pressure, but instead first builds trust in areas that are less directly tied to economic interests or political sensitivity, such as environmental data, telecommunications policy, and development planning, and waits for the right moment to make data requests. The partnership with the Ministry of Environment is a successful example. At the outset, the ministry was cautious about ODC's requests. Through sustained engagement over time, the ministry came to recognize that ODC handled released data responsibly and in the public interest. As a result, the ministry became more willing—under specified conditions—to share environmental impact assessment reports. Although not all reports can be made public, at least a channel for dialogue and negotiation has been established.

This difference reflects an important reality: the sensitivity of a data topic is closely tied to an agency's willingness to open it, and promotion strategies must be adapted to local conditions. From the German Commons' experience, this study observes that even in legally favorable environments with relatively clear data licensing, strategic topic selection and data curation are necessary to ensure dataset quality and usability. ODC's experience further reminds us that within nascent regulatory landscapes, topic selection is not merely a technical issue but also a matter of building trust and facilitating collaboration. Starting with less sensitive domains and gradually establishing successful cases often provides an effective pathway to unlock access to additional data sources.

4.2.4 Summary

Effective collaboration is key to the success of open data initiatives and AI applications. Whether within government agencies or in partnership with civil society organizations, establishing trust and shared understanding is essential. This study identifies the following factors for successful collaboration:

1. Technical capability is important. But understanding needs and effective communication are equally indispensable.

2. A bottom-up, demand-driven approach can promote collaboration and enhance the practical relevance of outcomes.

3. Adjusting strategies according to institutional environments and data sensitivity helps to address challenges. Institutional design and flexible collaboration models are complementary, and both can produce tangible results.

For government agencies considering the integration of open data and AI, the message from these experiences is clear: collaboration is a necessary condition for success—whether it is cross-agency cooperation within the public sector or public–private partnerships among stakeholders. Sincerely understanding each other's needs and constraints, and finding shared values and goals on that basis, is essential. Once mutual understanding and trust are established, data can flow more smoothly, technology can be applied more effectively, and broader socio-economic benefits become more achievable.

Chapter 5: Conclusion and Roadmap for Promoting AI-Ready Open Data

This chapter draws on the concrete experiences accumulated from the five case studies, sequentially outlining the technical choices for format conversion, practical methods for personal data de-identification, identification and handling of training data biases, and the planning of long-term data management infrastructure. Each topic presents the tools used in the cases, the limitations encountered, and the considerations behind selecting specific approaches, focusing on actionable steps that frontline practitioners can apply. Some agencies or teams, motivated by professional responsibilities or service quality considerations, may wish to invest extra effort in data processing. This chapter serves as a practical guide for those readers.

5.1 Implementation Modules and Roadmap

Promoting government data toward AI-ready is not a linear, one-way process; rather, it consists of a series of actionable modules that can be deployed according to the needs of each agency. In practice, individual agencies frequently encounter multifaceted and concurrent challenges. For example, an agency may have strong hardware infrastructure yet encounter institutional ambiguity regarding licensing terms, or face operational hurdles in optimizing data interoperability for AI while fostering strategic partnerships. The pace of development can also vary across agencies: some departments may advance more expeditiously due to the nature of their work or organizational culture, while others require more time to establish trust. Based on international case studies, this research categorizes these successful outcomes into four operational modules. Agencies can dynamically combine these practical recommendations according to the challenges they are currently facing, rather than adhering to a rigid, linear implementation path.

Roadmap to Tacking Challenges of Establishing AI Ready Data

Building Collaborative Relationships

Steps to Be Taken	Assess of institutional requirements and identify of prospective partners	Provide technical training to build capacity	Demonstrate the value of technical support
Actionable Modules	• Start with less sensitive areas, such as environmental data or communication policies. • Use workshops to directly ask civil servants about the difficulties they encounter and provide solutions for specific pain points. • Choose units that are already seeking assistance as the first partners, rather than the most influential agencies.	• Collaborate with local educational institutions to offer data literacy courses for civil servants. • The training should cover practical skills such as how to use data collection tools, principles of data organization, and data quality management. • Emphasize how these skills can improve daily work efficiency, rather than focusing solely on the concept of open data.	• Direct technical facilitation to address institutional data governance hurdles, such as the digitization of physical archives, creating data catalogs, and the optimization of data acquisition instruments. • This allows agencies to see that the partner is not adding extra work, but genuinely helping to reduce their workload.
Reference Cases	• In Cambodia, ODC started with the Ministry of Environment, assisting in organizing dispersed environmental impact assessment reports. • In Berlin, Germany, CityLAB identified through consultative workshops that the manual processing of parliamentary inquiries constituted a significant operational bottleneck and developed the Parla system	• ODC collaborated with the Cambodian Academy of Digital Technology to train staff from the Ministry of Planning and other relevant government ministries in by using data collection tools. • In Spain, ImpulsaDATA provided data governance training to various ministries, helping agencies understand how proper internal data management can enhance work efficiency.	• ODC assisted the Ministry of Environment in digitizing physical environmental impact assessments, enabling the scattered documents to be effectively managed. • The trust built during this process later encouraged the ministry to share more data when conditions allowed.
Expected Outcomes	• Establish at least one successful case to demonstrate the tangible value of collaboration to other agencies.	• Civil servants acquire basic data management skills and gain an understanding of the importance of data standardization and documentation.	• The agency shifted from evaluation to openness, becoming willing to discuss further possibilities for collaboration.

Clarify Licensing Terms

Steps to Be Taken	Adopt internationally standardized licensing terms	Refer to international practices in the same field	Establish a licensing decision-making process
Actionable Modules	• Scientific data often use CC BY 4.0, while government statistical data can adopt CC0 directly. • The key is not to adopt the most permissive license, but to select a clear and widely understood one.	• Develop internal guidelines specifying which types of data correspond to which licensing terms. • When institutional ambiguity exists regarding licensing, adopt incremental liberalization, contingent upon institutional confidence, while formalizing the rationale for licensing determinations to consolidate institutional knowledge.	• Create a single access point where users can consolidate multi-agency data access, with each dataset clearly labeled with its licensing terms and standardized metadata. • The platform should enforce fundamental format interoperability so that users do not need to handle differing data formats from each agency.
Reference Cases	• The licensing practices of open data platforms in various countries. Taiwan' s Open Government Data Platform also provides government data open licenses as selectable options.	• Spain' s ImpulsaDATA provides institutional guidance to agencies in conducting diagnostics to clarify which data can be safely opened and which licensing terms should be applied.	• The French open data initiative platform (data.gouv.fr) centralizes datasets released by various ministries, allowing the CroissantLLM team to directly download the needed data from a single access point without having to request authorization from each agency individually.
Expected Outcomes	• Establish a predictable licensing environment to reduce users' concerns about legal risks.	• Agencies have a clear basis for selecting licensing terms internally, reducing decision-making time.	• This significantly reduces the acquisition cost for data users and increases the likelihood that the data will be utilized.

Establish Data Circulation Mechanisms

Steps to Be Taken	Establish a centralized data platform	Institutionalize open data within standard operating procedures	Harmonization of regulatory frameworks	Provide companion-style support
Actionable Modules	• Creat a platform as a signal access point point where users can consolidate multi-agency data access, with each dataset clearly labeled with its licensing terms and standardized metadata. • The platform should enforce fundamental format interoperability so that users do not need to handle differing data formats from each agency.	• When an agency finalizes an official publication or statistical dataset, it should expeditiously ingest disclosable data segments into the centralized platform, instead of adhering to a reactive disclosure model. • This proactive approach turns open data from a discretionary task to a core institutional function.	• Assess the inter-legal dynamics among the Freedom of Government Information Law, the Personal Data Protection Act, and the Copyright Act. • Clarify the hierarchy of legal precedence and implement regulatory adjustments to ensure legal coherence and mutual reinforcement. Establish an inter-ministerial coordination mechanism to discuss concrete cases of legal application.	• During the institutionalization phase, comprehensive engagement with individual agencies remains imperative to deliver bespoke implementation roadmaps. • This facilitates the identification of idiosyncratic institutional constraints, favoring an inclusive advisory model over top-down regulatory imposition.
Reference Cases	• The French open data initiative platform (data.gouv.fr) centralizes datasets released by various ministries, allowing the CroissantLLM team to directly download the needed data from a single access point without having to request authorization from each agency individually.	• In France, ministries handle data release at the same time as they complete their regular work, rather than doing it afterward. In Taiwan, some agencies have already listed the regular updating of open data as part of their official duties.	• Cambodia is discussing the operationalization of a unified data governance framework, with the goal of enabling agencies to adhere to harmonized structural specifications for data curation.	• Spain' s ImpulsaDATA uses a methodology of diagnosis, gap analysis, and action to help agencies identify which data is eligible for secure dissemination. This consultative model fosters greater institutional buy-in compared to the unilateral imposition of standardized mandates.
Expected Outcomes	• This significantly reduces the acquisition cost for data users and increases the likelihood that the data will be utilized.	• Data can flow into the platform continuously and steadily, rather than depending on the initiative of individual staff members.	• Agencies have a clear understanding of how regulatory frameworks apply, thereby mitigating non-disclosure risks stemming from legal ambiguity.	• Systematic implementation mitigates institutional friction resulting from a lack of organizational contextualization, allowing each agency to progress at its own pace.

Pursuing the Highest Data Quality: Building the Technical Infrastructure

Steps to Be Taken	Refer to international practices in the same field	Establish a licensing decision-making process	Establish a centralized data platform	Establish a quality monitoring mechanism
Adopt internationally standardized licensing terms	• Scientific data often use CC BY 4.0, while government statistical data can adopt CC0 directly. The key is not to adopt the most permissive license, but to select a clear and widely understood one.	• Develop internal guidelines specifying which types of data correspond to which licensing terms. When institutional ambiguity exists regarding licensing, adopt incremental liberalization, contingent upon institutional confidence, while formalizing the rationale for licensing determinations to consolidate institutional knowledge.	• Create a single access point where users can consolidate multi-agency data access, with each dataset clearly labeled with its licensing terms and standardized metadata. • The platform should enforce fundamental format interoperability so that users do not need to handle differing data formats from each agency.	• Set data quality standards covering aspects such as completeness, accuracy, and timeliness. • Establish regular inspection processes instead of checking only at the time of release. Record user feedback as a basis for improving data quality.
Refer to international practices in the same field	• The licensing practices of open data platforms in various countries. Taiwan' s Open Government Data Platform also provides government data open licenses as selectable options.	• Spain' s ImpulsaDATA provides institutional guidance to agencies in conducting diagnostics to clarify which data can be safely opened and which licensing terms should be applied.	• The French open data initiative platform (data.gouv.fr) centralizes datasets released by various ministries, allowing the CroissantLLM team to directly download the needed data from a single access point without having to request authorization from each agency individually.	• In Berlin, Germany, Parla performs extensive manual checks to ensure quality when processing OCR conversions. • They compiled their experience into technical documentation and shared it publicly, so that others can avoid making the same mistakes.
Expected Outcomes	• Establish a predictable licensing environment to reduce users' concerns about legal risks.	• Agencies have a clear basis for selecting licensing terms internally, reducing decision-making time.	• This significantly reduces the acquisition cost for data users and increases the likelihood that the data will be utilized.	• Continuously improve data quality to build users' trust in the data.

This table is designed to help readers quickly identify their own situation and find relevant reference cases. It is important to understand that each case has found feasible solutions within its own constraints, and these solutions are not mutually exclusive—they can be referenced and flexibly combined.

It is not necessary to follow every step exactly. What matters more is that these solutions are interdependent. The trust built through collaboration is the foundation for subsequent systematic implementation. Companionship mechanisms developed to overcome obstacles can help more agencies cross initial thresholds. Technical experience accumulated for pursuing high-quality data can be shared back with implementers as a reference. This flexible yet robust approach often handles the complexities of real-world environments better than strictly adhering to a predefined roadmap.

5.2 AI-Ready Data Empowering Democracy and Society

When discussing the integration of open data with AI, it is easy to get caught up in purely technical debates and overlook the the mroe fundamential questions for civil servants, citizens, and stakeholders other than business sectors: “What is all this for?” From the cases examined in this study, we identified three dimensions of empowerment, which together form the foundational value of open data in contemporary society.

5.2.1 Empowering Civil Servants in Their Work

The most direct beneficiaries of open data initiatives are ultimately the civil servants themselves. The experience of the Parla project provides a robust empirical illustration of this principle. Consultative workshops with Berlin’s administrative personnel identified the manual processing of parliamentary inquiries as a significant operational bottleneck, the solutions developed by CityLAB not only helped them save time but, more importantly, enabled them to respond more effectively to members of parliament, thereby improving the quality of their work.

Spain' s ImpulsaDATA program reflects the same value. In the interview, Carlos Alonso Peña, Director of the Spanish Data Office Division, mentioned that many agencies, after receiving guidance, found that establishing good internal data governance not only facilitates external data release but also improves internal workflows. When data is properly organized and managed, staff can handle information retrieval, report preparation, and responses to tasks assigned by superiors much more efficiently. These internal benefits often serve as a key motivation for agencies to continue investing in open data initiatives.

The data literacy training that ODC provides to Cambodian government agencies illustrates the same logic. By helping agencies understand how to organize and manage data, ODC not only lowers the barriers to data release but also enhances the agencies' administrative effectiveness. This experience demonstrates that promoting open data should not be seen as an additional burden, but rather as an opportunity to strengthen the overall capacity of the public service system.

5.2.2 Empowering Citizens’ Knowledge and Participation

The contribution of open data to democracy goes beyond increasing transparency; it also lies in how it enables citizens to participate more effectively in public affairs. A typical example is the economic land concession map created by ODC in Cambodia. Before this map existed, indigenous communities had difficulty knowing which lands had been designated as concessions and which areas were still available for community use. Once this information was visualized, communities could concretely advocate to the government for the protection of their traditional territories. When such data is released in an AI-ready format, even citizens without specialized legal or spatial analysis backgrounds can potentially feed structured data into large language models to expeditiously summarize the potential impacts of specific development projects on their traditional territories, or even request AI to generate discussions about these projects.

This form of empowerment helps translate transparency from a principle into practical civic agency. When citizens have access to structured data, they can conduct their own analyses, formulate evidence-based inquiries, or develop alternative solutions. Data-driven civic participation is often more persuasive than non-empirical or purely rhetorical advocacy and is more likely to drive substantive policy change. Such empowerment shifts democratic engagement from passive awareness to active involvement. With well-structured raw data, citizens can use tools like ChatGPT to analyze information, transforming originally dense government budgets or meeting records into accessible, plain-language insights.

Most importantly, combining AI with open data lowers the expertise barrier to public participation. Previously, only senior researchers or large interest groups had the resources to conduct large-scale data analysis. Now, any citizen with basic questioning skills can leverage AI assistance to engage in evidence-based dialogue with the government. When open data transitions from passive digital information to AI-interoperable content, enabling automated synthesis and analysis, it not only strengthens citizens’ right to information but also enhances the efficacy of transparent oversight mechanisms regardless of resource limitations, reshaping the balance of power between citizens and government.

5.2.3 Empowering AI with Cultural Diversity

When viewed from a global perspective, the significance of open data for AI development goes beyond a purely technical issue; it touches on fundamental matters of cultural preservation and sovereignty. The goal of the CroissantLLM project is to build a truly bilingual French model, because the team observed that models claiming to support French often only include a small amount of French data within predominantly English training corpora, resulting in performance on French tasks that lags far behind their English capabilities.

The seriousness of this issue lies in the fact that if a country's language is underrepresented in training data, its citizens risk being marginalized or misinterpreted when using mainstream AI models. Models tend to output values and perspectives present in dominant training datasets. If historical events are deficient in localized linguistic nuances, models may rely on foreign media or biased minority sources to interpret history, leading to distorted public understanding of their own past.

ODC's experience further highlights the complexity of handling linguistic diversity. Staff members noted in the interviews that Cambodia's language challenges extend beyond Khmer and English to include numerous indigenous languages. If data are concentrated only on urban or dominant languages, the voices of rural communities and indigenous peoples risk disappearing from the digital sphere. This cultural-level inequality is likely to become even more pronounced as AI applications become more widespread. In this context, the role of government is particularly crucial. When private enterprises tend to collect data with commercial value, governments bear the responsibility to release data that may seem niche but possess high public value—such as local gazetteers, dialect records, or infrastructure data from remote areas. Although these data may not be commercially profitable, they are an essential piece in correcting cultural biases in AI models and building a digital environment that preserves local subjectivity.

The German Commons project, which faced challenges of temporal bias, offers a similar lesson. Because copyright restrictions mean that available cultural works are largely historical documents, models trained on them may adopt outdated language and phrasing. While this problem cannot be fully resolved within existing legal frameworks, it serves as an important reminder: how should governments adjust copyright policies or data release strategies to ensure that contemporary culture is also learnable and preservable by AI models?

5.2.4 Towards a Sustainable Open Data Ecosystem

When civil servants realize that open data can improve their work efficiency, they become more willing to engage in these efforts. When citizens can effectively use open data to participate in public affairs,[18] the government experiences positive feedback from data release. And when AI models can more accurately understand and generate local languages and cultures, society as a whole benefits from more appropriate and effective digital services.

Carlos Alonso Peña, Director of the Spanish Data Office Division, highlighted a particularly important point about cultural change. He noted that the greatest challenge in promoting open data is not technical or legal, but organizational culture. When civil servants fear making mistakes or worry that the quality of data is insufficient and might be criticized, open data initiatives stall. However, if the narrative is reframed so that agencies understand that open data is not just about external oversight but also about internal improvement, enabling better citizen participation, and ensuring that local culture is not marginalized in the AI era, the resistance to implementation is greatly reduced.

From this perspective, integrating open data with AI is not merely a technical upgrade. It is about whether civil servants can serve citizens more effectively, whether citizens can genuinely participate in public decision-making, and whether a country's culture and language can maintain subjectivity in the digital age. These three dimensions of empowerment together form the deeper significance of promoting open data and represent the core purpose that should always be remembered amid various technical and institutional challenges.

5.3 Research Limitations and Future Outlook

This study was conducted between August and December 2025. During this period, governments, civic organizations, and academic institutions worldwide continued to release more AI-ready datasets based on open data, as well as publish new policies and research findings. Due to the constraints of the project timeline, it is possible that the collection of cases and the study itself may not be fully comprehensive, and readers' understanding is appreciated. Most of the case studies in this report were selected and presented based on the project inventory in Chapters Two and Three and the TOE quadrant blueprint. However, as noted above, AI development across countries is rapidly evolving, and the classification of cases within the analytical matrix remains inherently dynamic. Due to time limitations, this report can only reflect the situation for the majority of the

18. Kleiman, Fernando, Marijn Janssen, Sebastiaan Meijer, and Sylvia J. T. Jansen. "Changing Civil Servants' Behaviour Concerning the Opening of Governmental Data: Evaluating the Effect of a Game by Comparing Civil Servants' Intentions before and after a Game Intervention." International Review of Administrative Sciences, 2022. https://doi.org/10.1177/0020852320962211.

project period. Additionally, cases in the fourth quadrant—characterized by high technological development but low organizational and environmental maturity—remain relatively scarce.

Finally, this study was conducted by a Taiwanese research team, and during the process, we interviewed six experts from the public sector, civic tech communities, and AI data training communities. These experts suggested several Taiwanese cases for potential further investigation. For example, in September, the Ministry of Digital Affairs announced an AI-ready metadata framework, providing Taiwan's government agencies with more opportunities to improve data quality and its documentation. Additionally, at the end of December, the Ministry officially released Taiwan Sovereign AI Corpus. During the same period, Switzerland also released its own sovereign AI dataset. These cases represent opportunities for the team's future research. At the same time, Taiwanese government agencies have suggested that, for domestic cases, further integration of local contexts and dialogue with civil servants could be achieved by developing additional operational toolkits or checklists highlighting key aspects of certain cases. This is also the direction our team hopes to pursue in future dialogues.

5.4 Conclusion

This report hopes that by documenting these cases, it can serve as a mirror to help public sector workers understand their own roles. Through these writings and the close-to-practice international examples, each agency and every officer can recognize their current position and understand that, even when still in the early stages or facing obstacles, their efforts carry public value. Open data and AI are not solely the domain of technical units or early adopters; they are a collective endeavor in which everyone involved in public service can gradually participate and steadily contribute.

At the same time, this report is intended to serve as a "communication material" between the government and civil society. For NGOs and civic tech communities, when engaging with government agencies of varying maturity levels and knowledge backgrounds, it provides concrete examples of "how to get started," "how others have progressed," and "which paths are feasible and adaptable." Such dialogue moves beyond abstract advocacy and is grounded in practical experience and actionable steps.

Future researchers and policymakers are invited to view these cases as a starting point for ongoing dialogue. As AI technologies, regulatory environments, and societal expectations continue to evolve, governance models for open data will inevitably need continuous adjustment and redesign. Only by establishing long-term collaboration and feedback mechanisms among government, academia, and civil society can this ecosystem grow sustainably. The more people can see their own roles along this path and understand each other's constraints and expertise, the more open data can truly become a sustainable public infrastructure. This is not the responsibility of any single party, but a collective effort that must be shared and is worth achieving together.